



3468 - Measuring the Hubble constant with the next multiple-imaged lensed supernova

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
imaging				
	1	imaging visit	NIRCam Imaging	(1) AT2025mkn
	51	imaging visit	NIRCam Imaging	(1) AT2025mkn
spectroscopy				
	2	position 1	NIRSpec IFU Spectroscopy	(1) AT2025mkn

ABSTRACT

Spectroscopic time-delay measurements of multiply-imaged supernovae offer a very efficient way to measure the expansion rate of the universe. NIRCam and NIRSpec follow-up observations with JWST upon the next lensed supernova (SN) discovered by the Zwicky Transient Facility (ZTF) could pave the way for a novel technique to measure the Hubble constant and address the "Hubble tension". Unlike any other time-delay measurement, spectroscopic dating of multiple SN images can be done without repeated observations. We propose to select a suitable lensed survey for ToO observations and anticipate a 5% (or better) measurement of H_0 , with only 3 hours of JWST time.

OBSERVING DESCRIPTION

We propose to carry out non-disruptive ToO observations of a strongly lensed, multiply-imaged, supernova discovered by the ZTF. Thanks to the unique combination of near-IR sensitivity, throughput and angular resolution, JWST observations will allow us to measure positions and the spectroscopic time-delays between resolved multiple images, the latter with an accuracy of about half a day. We will accomplish this through SED dating of the individual resolved SN images, with the aim to establish a novel and powerful technique to measure the Hubble constant with strongly lensed supernovae. The requested observations will allow us to measure H_0 with 5% accuracy or better. This is particularly timely given the "Hubble tension", arguably the hottest issue in cosmology today.

Proposal 3468 - Targets - Measuring the Hubble constant with the next multiple-imaged lensed supernova

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	AT2025mkn	RA: 16 42 11.4696 (250.5477900d) Dec: +55 31 2.46 (55.51735d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This is the position of the bright transient image detected from the ground Category=Star Description=[Type Ia supernovae] Extended=YES				

Proposal 3468 - Observation 1 - Measuring the Hubble constant with the next multiple-imaged lensed supernova

Observation	Proposal 3468, Observation 1: imaging visit									Mon Jul 28 16:00:11 GMT 2025	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	AT2025mkn	RA: 16 42 11.4696 (250.5477900d)			Epoch of Position: 2000					
			Dec: +55 31 2.46 (55.51735d)								
			Equinox: J2000								
	Comments: This is the position of the bright transient image detected from the ground										
	Category=Star										
	Description=[Type Ia supernovae]										
	Extended=YES										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module Gap				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		2		SMALL-GRID-DITHER				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F277W	RAPID	2	1	4	4	85.894	63898	
	2	F150W	F277W	RAPID	2	1	4	4	85.894	63898	
Special Requirements	Target Of Opportunity Response Time 21 Days, Carry-Over										

Proposal 3468 - Observation 51 - Measuring the Hubble constant with the next multiple-imaged lensed supernova

Observation	Proposal 3468, Observation 51: imaging visit									Mon Jul 28 16:00:11 GMT 2025	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 51:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	AT2025mkn	RA: 16 42 11.4696 (250.5477900d)			Epoch of Position: 2000					
			Dec: +55 31 2.46 (55.51735d)								
			Equinox: J2000								
	Comments: This is the position of the bright transient image detected from the ground										
	Category=Star										
	Description=[Type Ia supernovae]										
	Extended=YES										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module Gap				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		2		SMALL-GRID-DITHER				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F277W	RAPID	2	1	4	4	85.894	63898	
	2	F150W	F277W	RAPID	2	1	4	4	85.894	63898	
Special Requirements	Offset 121.0 arcsec, 35.0 arcsec										
	Target Of Opportunity Response Time 21 Days, Carry-Over										

Proposal 3468 - Observation 2 - Measuring the Hubble constant with the next multiple-imaged lensed supernova

Observation	Proposal 3468, Observation 2: position 1										Mon Jul 28 16:00:11 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	AT2025mkn	RA: 16 42 11.4696 (250.5477900d)			Epoch of Position: 2000						
			Dec: +55 31 2.46 (55.51735d)									
			Equinox: J2000									
		Comments: This is the position of the bright transient image detected from the ground Category=Star Description=[Type Ia supernovae] Extended=YES										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	1 AT2025mkn	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	141127	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140M/F100LP	NRSRAPID	12	7	false	true	NONE	4	28	3908.184	